

ConSynth-X

A Multi-Condition Synthetic Dataset for
Construction Site Computer Vision Under Extreme Field Conditions

Viet Huy Duong Ruoxin Xiong, Ph.D.

Kent State University

April 2026

The Problem

Goal: Provide a comprehensive dataset for construction site computer vision that enables **training and evaluating** models across diverse extreme field conditions.

Why this matters:

- ▶ Construction sites face rain, snow, fog, low light, glare
- ▶ Models trained on clean data fail dramatically in the field:

Condition	Detection mAP@0.5	Drop
Clear weather (original)	0.819	—
Rain / Snow	0.446	−46%
Nighttime	0.454	−45%

- ▶ No existing dataset covers multiple extreme conditions for construction
- ▶ Collecting real extreme-condition data is dangerous and expensive

→ We synthetically generate a multi-condition dataset to fill this gap.

Our Solution: ConSynth-X

We take existing construction photos and digitally add weather, night, fog, etc.

What we add	How	Images	In simple terms
Rain / Snow (style)	Neural Style Transfer	57K	Copy weather “look” from reference photos
Rain / Snow (diff.)	InstructPix2Pix AI	10K	AI edits image based on text prompt
Nighttime	CycleGAN-Turbo	23K	AI converts day photo to night
Fog	Koschmieder model	3K	Physics-based fog simulation
Smaller objects	FLUX.1 outpainting	2K	Expand image canvas, objects appear farther
Total augmented		95K+	

Applied to 2 datasets: **Construction Site 10k** (3 classes) + **SODA** (15 classes).

All original bounding box annotations are preserved in augmented images.

Visual Examples



Each row: same photo → Style Transfer → AI Diffusion + physics rain/snow overlay.

Prior Work & Our Contribution

Ding et al. (2024):

- ▶ UIA-YOLOv5 + Neural Style Transfer
- ▶ ExtCon: 506 real extreme images
- ▶ +8.21% mAP improvement
- ▶ Only 1 method (NST), no snow/night/scale

ConSynth-X (ours):

- ▶ 126,000+ images ($250\times$ larger)
- ▶ 3 complementary pipelines
- ▶ 5 condition categories
- ▶ 5-dimensional validation framework
- ▶ Open-source pipeline for new datasets

Key question: How do we know the synthetic images are good enough?

→ We validate from 5 different angles.

How We Validate: 5 Questions

Each approach answers a different question about image quality:

1. **Weather Recognizability** — Can you tell it's raining/snowing?
2. **Distributional Fidelity** — Is it statistically similar to real weather photos?
3. **Texture Fidelity** — Are surface details (concrete, metal, fabric) preserved?
4. **Embedding Distance** — In AI “feature space,” is it close to real weather?

+ **Supplementary:** AI-generated image detection (UnivFD) — confirms our images retain real-photo statistics, but this metric is **not informative** for edit-based augmentation (see backup slide).

No single metric tells the whole story.

Cross-validation across all 4 reveals trade-offs between pipelines.

Validation 1: Weather Recognizability — Can You Tell It's Raining?

Question: Is the weather effect actually visible?

(An image can look “real” but not look like “rain.”)

Method: Pre-trained **weather classifier** (SigLIP2) classifies each augmented image into 5 weather classes.

Caveat: Classifier trained on driving scenes — it misclassifies 58% of *original clear* construction images as “rain” (cross-domain bias). **Relative comparison** between pipelines is more meaningful than absolute numbers.

Condition	Accuracy
Original (clear)	6% sun
ST Rain A	89%
ST Snow B	95%
Diff. Rain	64%
Diff. Snow	26%

Takeaway:

Style transfer → strong, obvious weather.

Diffusion → subtle, less recognizable.

Diffusion snow only 26%, but its texture quality is the best (see Validation 4).

Validation 2: FID/KID — Is It Statistically Similar to Real Weather?

Question: How close is the distribution of our synthetic images to the distribution of *real* weather photos?

Method: FID (Fréchet Inception Distance, Heusel et al. 2017) compares feature distributions between two image sets. Lower = more similar.

How: Compare augmented images against 3 real-weather datasets (ACDC, WeatherBench, WeatherNet).

“Baseline gap” = how far original clear images are from real weather. If augmentation *reduces* this gap, it’s working.

	FID	Δ
<i>Night (vs ACDC night):</i>		
Original	270.5	—
CycleGAN	178.9	−34%
<i>Snow (vs ACDC snow):</i>		
Original	216.0	—
Diffusion	205.3	−5%
<i>Rain (vs ACDC rain):</i>		
Original	189.8	—
ST rain	192.3	≈0%

Night is the biggest win:
34% closer to real night photos.

Validation 3: Texture Fidelity — Are Surface Details Preserved?

Question: Does augmentation damage the fine texture of surfaces (concrete, metal, fabric)?

Method: Adapted from **Duminil et al. (2025, Sci. Reports)**. Extract 4 texture features per image (GLCM, LBP, DCT, Haralick), compare distributions via Wasserstein distance, then fuse with **Dempster-Shafer belief theory**: all 4 features “vote” on texture quality.

H = belief “faithful” (preserved). $\bar{H}$ = belief “artifact” (damaged).

	H	$\bar{H}$	
Diff. Snow	.86	.00	Faithful
Diff. Rain	.85	.00	Faithful
ST Rain	.00	.26	Uncertain
ST Snow	.00	.83	Artifact
Fog	.00	.83	Artifact
Night	.02	.73	Artifact

Diffusion preserves texture
(all 4 criteria agree: $H > 0.85$).

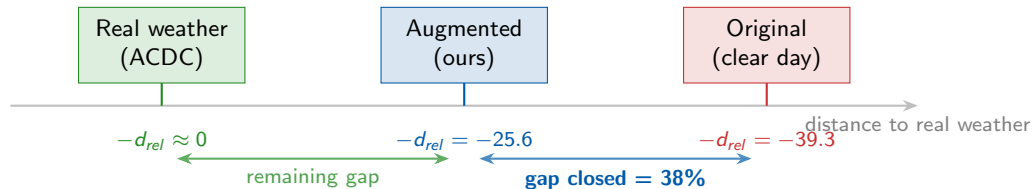
Style transfer creates visible weather but damages texture.

Night has highest “conflict”:
local texture OK, global lost.

Validation 4: Embedding Distance — The Idea

Question: In AI “feature space,” are our augmented images closer to *real* weather photos?

Method: Relative Mahalanobis Distance (Ruck et al. 2026). Subtract background to isolate weather signal.



Example: Night (CLIP). Original is -39.3 from real ACDC night. CycleGAN augmentation moves it to -25.6 — closing 38% of the gap.

$$\text{"\% gap closed"} = \frac{\text{original} - \text{augmented}}{\text{original} - \text{baseline}} \times 100$$

Validation 4: Results & Cross-Domain Limitation

CLIP (semantic/style)

Condition	Best method	Gap closed
Night	CycleGAN	38%
Snow	ST heavy	27%
Rain	ST moderate	25%
Fog	Physics	18%

All conditions move closer
to real weather in CLIP space.

CLIP captures “weather look”
(style/semantics) which is relatively
domain-agnostic: rain looks like rain
whether on a road or a construction site.

DINOv2 (texture/structure)

Condition	Best method	Gap closed
Night	CycleGAN	8%
Snow	IP2P	0.4%
Rain	IP2P	0.8%
Fog	Physics	5%

Near-zero movement.

But this is a **metric limitation**:

DINOv2 captures texture/structure.
Reference is ACDC (*driving* scenes) — wet
roads $\neq$ wet concrete. The **cross-domain
structural gap** overwhelms the weather
signal.

Takeaway: CLIP results validate our augmentation. DINOv2 gap reflects **domain mismatch**
(driving vs. construction), not augmentation failure. A construction-specific weather reference
would be needed — but that is exactly what does not exist yet.

Putting It All Together: The Trade-Off

What we measured	Style Transfer	Diffusion	Night
1. Weather visible? (classifier $\uparrow$)	62–95%	26–64%	N/A
2. Close to real weather? (FID $\downarrow$)	192–228	194–219	179
3. Texture preserved? ($H \uparrow$)	0.00	0.86	0.02
4. Embedding shift? (CLIP $\uparrow$)	25–27%	18%	38%

The key finding:

Style Transfer and Diffusion are **complementary**:

- ▶ Style Transfer: strong weather, damages texture
- ▶ Diffusion: natural texture, subtle weather

Neither is “better” — they serve different purposes.

Practical recommendation:

- ▶ **Training data:** use Diffusion (natural texture helps model learn)
- ▶ **Robustness testing:** use Style Transfer (extreme weather stress-tests the model)
- ▶ **Night:** best overall domain shift

Downstream Detection: Does Augmented Data Help?

Question: If we add synthetic weather images to training, does it improve or hurt the model?

Setup: YOLOv8n detector trained with/without augmented data, tested on clean images.

Training data	mAP@.5	mAP@.5:.95
Original only (7K)	0.562	0.338
+Weather (17K)	0.547	0.339
+Weather strict	0.538	0.349

Result: Adding 10K+ synthetic images does **not degrade** clean-condition detection (mAP@.5:.95 neutral ± 0.001).

This means synthetic quality is sufficient for training — the model doesn't get “confused” by augmented images.

Next step: Test on weather/night conditions (experiment ready, awaiting compute).

This will show the *robustness benefit*.

Paper Status & Next Steps

Target: Nature Scientific Data

Completed:

- ▶ 5 augmentation pipelines
- ▶ 5-dimensional validation
- ▶ 23-page manuscript, 21 references
- ▶ Prior work comparison (Ding et al.)
- ▶ Cross-validation synthesis

Review feedback:

“Underlying research has merit... could reach Minor Revision quality”

Remaining:

1. Cross-condition detection results
(experiment script ready, ~2h GPU)
2. SODA 15-class validation
3. Dataset upload (HuggingFace)
4. Final paper polish

Roadmap:

Paper 1 (Dataset) → Paper 2 (VLM Benchmark)

→ Paper 3 (Detection Improvement)

Backup: Why Not Use AI-Generated Image Detectors?

Two different detection problems:

SID (Synthetic Image Detection):

Classify *entire* image as real vs AI-generated.

Examples: UnivFD, DIRE, NPR.

Designed for: Midjourney, DALL-E,

StyleGAN outputs.

IML (Image Manipulation Localization):

Detect *which regions* of a real image were edited.

Examples: TruFor, X-Edit, DiffForensics.

Designed for: splicing, inpainting, local edits.

Our images are neither:

ConSynth-X starts from **real photos** and applies **global edits** (weather, night).

→ SID methods see “real photo” (99.5% fooling)

because most pixels are original.

→ IML methods expect **local** edits, but weather is a **global** transformation.

Our approach: Instead of binary real/fake, we validate through **domain-specific metrics** — weather recognition, distributional fidelity, texture preservation, and embedding distance.

UnivFD results (>99.5% fooling) are reported as supplementary. Full results in paper appendix.

Thank You

Questions?

Viet Huy Duong — vduong1@kent.edu

Ruoxin Xiong — rxiong3@kent.edu

Kent State University